

Towards Measuring Key Performance Indicators of Semantic Business Processes

Branimir Wetzstein, Zhilei Ma, and Frank Leymann

Institute of Architecture of Application Systems, University of Stuttgart, Germany
`{firstname.lastname}@iaas.uni-stuttgart.de`

Abstract. Business Activity Monitoring (BAM) enables continuous, real-time performance measurement of business processes based on key performance indicators (KPI). The performance information is employed by business users but prior support from IT engineers is required for setting up the BAM solution. Semantic Business Process Management (SBPM) tries to minimize the needed support from IT staff throughout the business process lifecycle. In this paper we introduce a framework for BAM as part of SBPM. We show how performance measurement related activities can be integrated into the semantic business process lifecycle. KPIs are modeled by business analysts exploiting semantic annotations of business processes. KPI models are automatically transformed to IT-level event-based models and used for real-time monitoring using reasoning technology.

Keywords: Semantic Business Process Management, Business Activity Monitoring, Key Performance Indicator.

1 Introduction

Business Process Management (BPM) encompasses methods, techniques, and tools that allow organizing, executing, and measuring the processes of an organization. Recently, companies have increasingly been using Business Process Management software to manage their business processes [SF03]. Typically, the business process lifecycle begins with the business analyst analyzing business processes in the company and creating process models using a process modeling tool. When the process is to be automated, it is translated by IT engineers to a workflow model, which is run on a process engine. The process engine executes the workflow model by delegating the process tasks to human workers and automated IT applications.

For controlling the achievement of business goals in business processes and measuring process performance, business activity monitoring (BAM) technology enables continuous, real-time monitoring of processes based on key business metrics, also known as key performance indicators (KPI). Business people define KPIs based on business goals at process design time. These KPIs are then translated to IT monitor models by IT engineers. At process execution time, event-based BAM technology displays information on KPIs in dashboards and alerts responsible business managers in case of severe deviations from planned values.

Despite of increasing software support for BPM in general and BAM as part of BPM in particular, there is still a low degree of automation in the business process lifecycle [HLD+05]. In particular, there are substantial difficulties when it comes to bridging the gap between the business and IT views of business processes. One of the major problems is the translation of the high-level business process models and corresponding KPI models, which are modeled by a business analyst, to workflow models and monitor models on the IT level. The vision of Semantic Business Process Management (SBPM) is to close the Business-IT gap by using semantic technologies [HLD+05].

In this paper we introduce a framework for modeling and monitoring of KPIs in SBPM. Our goal is to increase the degree of automation in measuring of KPIs by enabling the business analyst to define the KPIs and then generate the IT artifacts needed for monitoring. The contribution of the paper is as follows. (i) We integrate the KPI management into the overall semantic business process (SBP) lifecycle by depicting how KPIs are modeled, implemented and monitored in the respective SBP phases. (ii) We create a KPI ontology which enables the business analyst to define the KPIs based on ontology concepts which are part of semantic annotations of semantic business processes. Semantic business processes explicitly specify the semantics of process activities by modeling their inputs, outputs, preconditions and postconditions, in terms of business objects (e.g., “purchase order”) and their state changes (e.g., “purchase order received”). As KPIs are also based on business objects (e.g., “percentage of purchase orders which were delivered successfully and on time”), semantic annotations of processes can be exploited for the definition of KPIs in the SBP modeling phase. (iii) We depict how the KPI model is transformed to a monitor model, which supports evaluation of KPIs based on events published by the process engine at process execution time. As the KPIs are defined on an ontological level, we show how reasoning technology is used for their computation.

The rest of the paper is organized as follows. In Section 2 we depict how our performance management framework fits into the overall SBP lifecycle. Section 3 describes how KPI models are created in the SBP modeling phase. Section 4 describes how KPI models are transformed to executable monitor models and how they are measured at process execution time. In Section 5 we present related work, and finally, in Section 6 we outline our future work.

2 Performance Management within the SBP Lifecycle

In [Fa07] the SBP lifecycle has been defined as consisting of four phases: SBP modeling, SBP configuration, SBP execution, and SBP analysis. Apart from these four phases of the SBP lifecycle, there is a business strategy definition layer which is providing the basis for SBPM by defining the business strategy and business goals for accomplishing that strategy. Business goals are broken down to key performance indicators which are key business metrics for measuring the business performance. By measuring the KPIs, responsible business people control whether the business goals are achieved as planned.

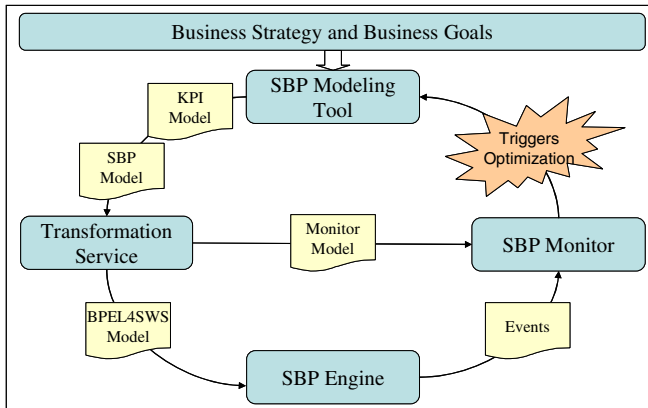


Fig. 1. Performance Management Lifecycle

In the first phase of the SBP lifecycle, the SBP modeling phase, business analysts model business processes and corresponding KPIs based on the requirements posed by the business goals in the strategy layer.¹ Business analysts use the SBP modeling tool to create a process model based on a process modeling notation such as BPMN [BPMN06] and annotate it semantically by referencing concepts from domain ontologies. These concepts describe business objects which are worked on in the business processes. The business analyst specifies the KPIs based on the business objects and their state changes. A KPI is related to the process model by referencing process activities and their semantic annotations. A KPI contains a computation definition and a target value which is to be accomplished to satisfy a business goal. In addition, allowed ranges of values and in case of deviations the resulting notifications can be defined. The resulting artifacts in the SBP modeling phase are an SBP model and a corresponding KPI model.

In the SBP configuration phase the SBP model is transformed to an executable workflow model, in our case BPXL4SWS [NLK+07], which is an extension of BPEL towards interaction with Semantic Web Services. The KPI model is automatically transformed to a monitor model which is an IT-level representation of the KPI model. In addition, deployment information for the SBP engine is generated. It specifies for which activities of the BPXL4SWS process model logging events should be published at process runtime. The SBP monitor subscribes to these events and evaluates the KPIs based on them at runtime. The executable BPXL4SWS process model is deployed to the SBP engine; the monitor model is deployed to the SBP monitor.

In the SBP execution phase, the SBP engine executes the process and publishes events to a publish/subscribe event infrastructure. The SBP monitor receives the events and computes the corresponding KPI values. As the KPIs are based on semantic descriptions, a reasoner is used for their evaluation. In case of deviations from allowed values, notifications are sent to responsible business users. The KPI values are displayed in dashboards.

¹ The relationship between business goals at the strategy layer and KPIs specified at the process modeling layer is out of scope of this paper.

Finally, in the SBP analysis phase, the KPI values are analyzed and the achievement of business goals is evaluated. In case of non-satisfying performance, causes are tried to be found, which eventually can lead to optimization of processes in the SBP modeling phase, thus closing the SBP lifecycle. The SBP analysis phase is out of scope of this paper.

3 Modeling Key Performance Indicators

KPIs are key business metrics for getting information on the performance of a company. In this paper we concentrate solely on KPIs which evaluate the efficiency and effectiveness of business processes and which can directly be derived and computed based on the runtime data of the executable business processes. Other KPIs, e.g. financial metrics such as Return on Investment (ROI) are not directly measurable by using runtime data of business processes and thus are out of the scope of this paper.

A KPI can be seen as a key process metric, which is evaluated in a certain analysis period and has a target value which should be reached or preserved within the analysis period indicating the achievement of predefined business goals [SS06, WAM+07]. Typically, in addition to a target value, allowed ranges of values are defined. In case of deviations, notifications are sent to business people in real-time. An example for a KPI is “percentage of purchase orders that were processed in time in the last 3 months should be greater than 90%”, the analysis period being 3 months and the target value 90%.

Our analysis of concrete KPIs, such as SCOR Supply Chain Metrics [SCOR07], and examples in related work [MR00, SS06], shows that most KPI computations are specified based on business objects which are dealt with in the process to be measured. Such business objects (a.k.a. business items, business documents or business data) are for example “purchase order”, “invoice”, or “shipment”. Typically, KPIs contain a condition on the state of these business objects or query their attributes, e.g. “number of purchase orders processed successfully and in time”.

Figure 2 shows two typical KPI templates, which are based on business objects [SS06]. For example, for a “purchase order process” the “business object” would be replaced by “purchase order”.

In order to be able to model KPIs for business processes, one has to know which business objects are handled by a process and which state changes occur at which

KPI	Computation Expression
Deadline Adherence	$\frac{\text{Number of finished Business Objects on time}}{\text{Number of all finished Business Objects}} \times 100\%$
Process Duration	$\frac{\sum (\text{finish date} - \text{start date}) \text{ of all finished Business Objects}}{\text{Number of all finished Business Objects}}$

Fig. 2. KPI Templates based on Business Objects

activities. Often, however, business objects and their state changes are not explicitly modeled as part of business process models in the process modeling phase. In most cases they are implicitly part of the names of activities, e.g. “Receive Purchase Order”. In semantic business processes, however, activities specify the semantics of their inputs, outputs, preconditions and postconditions²³ explicitly by referencing concepts defined in domain ontologies. These concepts model business objects and information on their state changes [BDW07]. Semantic business processes thus explicitly model at which activities business objects are processed and how their states change. Therefore, as the information needed for KPI specification is provided already in the process modeling phase, we can pursue a model-driven approach: the KPIs are specified by business analysts as a mathematical formula based on business objects as sketched in Figure 2 and then automatically transformed to an event-based monitor model which is used for their evaluation at process execution time. When it comes to tool support for the business analyst, besides providing support for entering a mathematical formula, the tool can present the business objects and the possible state changes which are part of semantic annotations of process activities automatically and thus help the analyst determine the objects and states he can use in the formula. Implementing support for specifying KPIs as part of an SBP modeling tool is part of our future work.

3.1 KPI Ontology

For the specification of KPIs we have defined a KPI ontology using WSML [BLK+05] as the ontology language.⁴ Figure 3 shows the KPI ontology and its relation to the relevant concepts from the process modeling ontology [He07] and the event ontology [PD07]. A KPI model groups KPIs for a business process. A KPI consists of an aggregate metric definition, which computes the actual value of the KPI in a certain analysis period. The target value of the KPI specifies a value which is to be achieved in the analysis period. For deviations from target values alerts can be defined.

When formulating metrics for KPIs we can distinguish between instance metrics and aggregate metrics. An instance metric calculates metrics for one single process instance of a process model. Instance metrics measure duration between two activities in the process or test a condition on the state of the process instance at a certain point in the process. The point in time when instance metrics are to be evaluated is modeled by execution events. Execution events are part of the event ontology and provide the needed link of the KPI computation to the process model. At process runtime these events are published by the process engine before and/or after the execution of

² Pre/postconditions denote the “state of the world” before and after the execution of the activity. In particular, a pre/postcondition can contain assertions which are not reflected in the input/output of the activity, i.e. as an attribute of a business object.

³ We have chosen technology-neutral terms here. In WSMO, the corresponding terms are: preconditions, postconditions, assumptions and effects.

⁴ When evaluating KPIs, we will not reason over the KPI ontology, but over domain ontologies which contain the business objects. Nevertheless, we have used an ontology language for the KPI metamodel because it is interlinked with the process ontology and event ontology, and all SBPM-related metamodels in our framework are defined as WSML ontologies [He07].

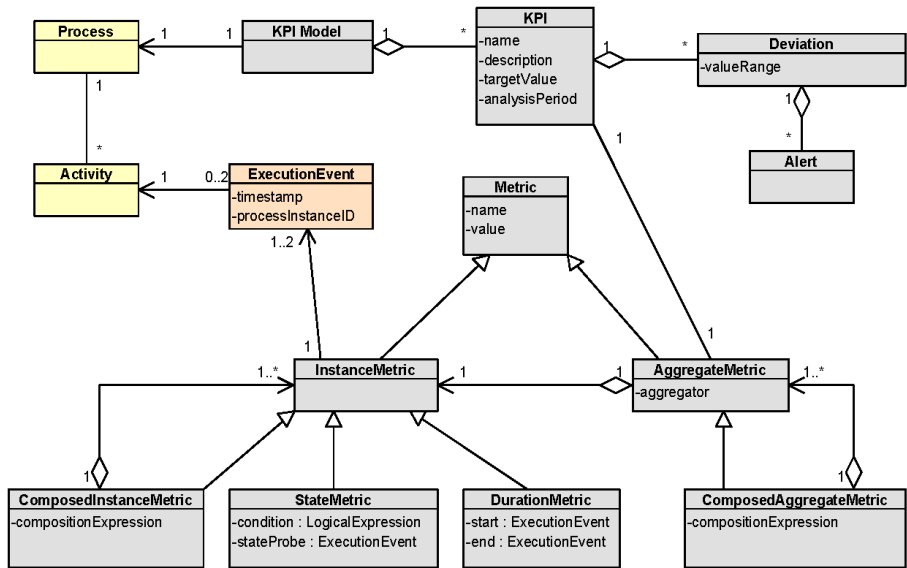


Fig. 3. KPI Ontology as a UML Class Diagram (simplified)

process activities and are received by the process monitoring tool for calculation of the KPIs. The execution events provide the attribute “timestamp” which enables the calculation of the duration between activities. The duration metric measures the time between two activities by simply subtracting the “timestamp” attribute values of two execution events.

A state metric references an execution event and specifies a logical expression which formulates a condition on the state of the process instance at that point in time. The state of the process instance at a certain point in time is given on the one hand by the values of the business objects, and on the other hand by assertions which are part of the postconditions but are not directly modeled as attributes of business objects; e.g. we could define a WSMML relation “OrderDeliveredSuccessfully(Order)” which acts as an assertion in the postcondition of an activity to express successful processing of a purchase order process. The state metric condition evaluates to true or false. The logical expression is formulated in the ontology language which is used for the specification of the domain ontologies, e.g. in WSMML, and constitutes thus an “intelligent query” of the state of the process instance. In addition to specifying duration and a state condition, instance metrics can be composed using arithmetic and logical operators. Thus, a duration metric could alternatively be defined as a subtraction of the execution event timestamps of two state metrics; we have defined an explicit duration metric concept for convenience reasons.

Aggregate metrics aggregate values of instance metrics using operators “max”, “min”, “average”, and “sum”. They thus evaluate the performance of a process in a certain analysis period. For example, if a state metric evaluates whether a “purchase order has been processed in time”, an aggregate metric would evaluate “the number (sum) of purchase orders processed in time” in a certain analysis period. Aggregate metrics can be composed using arithmetic operators.

An important aspect of KPI monitoring is the notification of business users in case of deviations from target values. We define deviations by specifying numerical ranges around the target value which the actual value is allowed to be in. In case of a deviation from this range of values an alert is raised, e.g., by sending an email, invoking a web service, or displaying a message in the dashboard.

3.2 Example

In the following we will demonstrate the use of the KPI ontology based on an example business process. Figure 4 shows a business process for processing purchase orders: After receiving a purchase order, the order is first analyzed. If it can be processed without any changes an acknowledgement is sent, otherwise a new offer is sent to the customer, resulting in a changed purchase order. In the next step the purchase order is processed, shipped, and finally a shipment receipt acknowledgement is received from the customer.

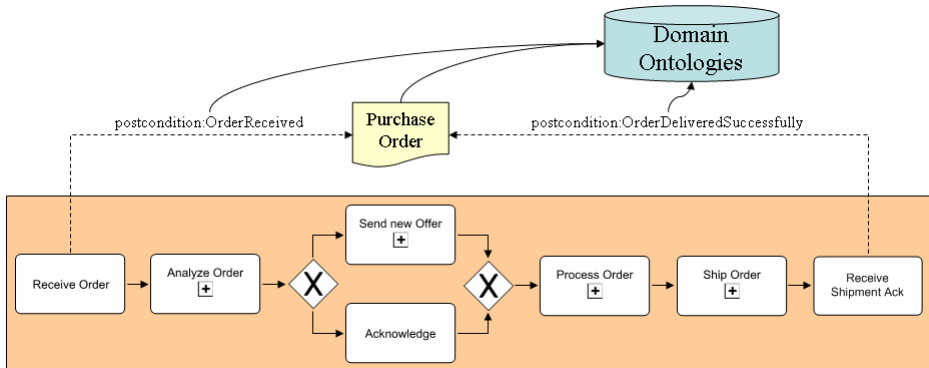


Fig. 4. Purchase Order Process

We define two KPIs which measure deadline adherence and process duration for that process (Figure 5). For the definition of the two KPIs we make following assumptions: the purchase ontology defines an “Order” concept, which contains an attribute “assuredDeliveryDate”; the purchase ontology defines (WSML) relations for the different states of the “Order” concept, e.g. “OrderReceived(Order)” and “OrderDeliveredSuccessfully(Order)”; these relations are used in postconditions of process activities, e.g. after the last activity either the relation “OrderDeliveredSuccessfully(Order)” or “OrderDeliveredLate(Order)” could hold; the purchase ontology contains additional relations which categorize orders based on products they contain, e.g. “ElectronicProductOrder(Order)” for orders that contain only electronic products.

For calculating the process duration we measure the time between the end and start activity by specifying a duration metric referencing the two corresponding execution events. An aggregate metric calculates then the KPI value by aggregating the duration metric values using the “average” operator.

We measure deadline adherence as the percentage of orders delivered successfully and on time whereby we are interested in orders that only contain electronic product items. The computation formula of that KPI (Figure 5) is mapped to five metrics: an instance state metric to evaluate whether a purchase order is processed successfully and on time, a corresponding aggregate metric which sums up the instance metrics which evaluate to “true”, an instance state metric which evaluates to “true” when a new purchase order is received, the corresponding aggregate metric which counts those occurrences, and finally, an aggregate metric which composes the two former aggregate metrics to compute the percentage value.

KPI	Computation Expression
Deadline Adherence	$\frac{\sum \text{ShipmentAckEvent.timestamp} \leq \text{Order.assuredDeliveryDate} \text{ AND OrderDeliveredSuccessfully(Order) \text{ AND ElectronicProductOrder(Order)}}{\sum \text{OrderReceived(Order)}} \times 100\%$
Process Duration	Avg (ShipmentAckEvent.timestamp – ReceiveOrderEvent.timestamp)

Fig. 5. KPI definitions for the Purchase Order Process

The benefit of using ontologies in our approach is shown when evaluating state metrics. In order to define a state metric which evaluates whether the purchase order for electronic products has been processed successfully and on time, we need to test a condition on the last activity of the process. We therefore define the execution event “ShipmentAckEvent” which occurs after the activity “Receive Shipment Acknowledgment” has been processed. For testing whether the purchase order was processed on time, we compare the “assuredDeliveryDate” with the timestamp of the execution event. For testing whether the purchase order has been processed successfully, we check whether the relations “OrderDeliveredSuccessfully(Order)” and “ElectronicProductOrder(Order)” hold. The WSMML Logical Expression for the instance metric looks as follows (namespaces omitted):

```
?order[assuredDeliveryDate hasValue ?plannedDate] memberOf Order
  and ?event[activity hasValue ?activity] memberOf ExecutionEvent
  and ?activity[name hasValue “Receive Shipment Acknowledgment”]
  and ?event[timestamp hasValue ?eventDate]
  and (dateLessThan(?eventDate, ?plannedDate)
    or (dateEqual(?eventDate, ?plannedDate))
  and OrderDeliveredSuccessfully(?order)
  and ElectronicProductOrder(?order)
```

The WSMML logical expression is automatically generated based on a formula, which the business analyst specifies with tool support (as sketched in Figure 5). It is then utilized as a query to the WSMML reasoner at process runtime (Section 4). The benefit of our approach is that implicit knowledge can be taken into account when

evaluating state metrics. In our example, based on the WSML relations, which are defined by appropriate WSML axioms, the reasoner would infer whether the order is an electronic product order based on a product ontology and evaluate the postcondition of the activity.

4 Monitoring Key Performance Indicators

After the semantic business process and KPIs have been modeled in the SBP modeling phase, they are transformed to executable artifacts in the SBP configuration phase. This involves the transformation of the SBP model to an executable BPEL4SWS process model, and the transformation of the KPI model to a monitor model for the SBP monitor and SBP engine specific deployment information on which events have to be published at runtime.

When KPI models are specified for an SBP model, in addition to the generation of the executable BPEL4SWS process model, information is needed for specifying how the execution events which are bound to activities and are used for the evaluation of instance metrics are published at process execution time. There are basically two approaches how to generate these events: (i) The generated BPEL4SWS process model does not contain any event specific logic, i.e. events are transparent to the BPEL4SWS process model. A BPEL engine typically already provides a mechanism for logging the execution of process instances by using events based on an internal event model. These events are stored in an audit log. Additionally, events can be published on a publish/subscribe infrastructure to enable real-time monitoring, which is what is required in our case. As only a small fraction of possible events is needed for KPI evaluation, deployment information for the SBP engine is generated which specifies for which activities events are to be published at runtime. This information is derived from the execution event definitions in the KPI model; the deployment information is specific to the SBP engine implementation, as the BPEL specification does not specify an event model. In our case the events are execution events defined in the event ontology and contain instance data of business objects which are defined in domain ontologies. The SBP engine is adapted to generate this kind of events. (ii) The generated BPEL4SWS process model contains explicit activities which publish events to a predefined interface provided by the SBP monitor. The events contain the specific data which is needed for the computation. The event data cannot however contain technical information such as the time of the activity execution. Also the BPEL4SWS process now contains much technical information which makes it less readable. Therefore, we have chosen the first approach.

The KPI model is transformed to a monitor model. The monitor model specifies how events received by the SBP monitor are processed to calculate the KPIs. For event handling, complex event processing (CEP) technology could be used [Lu02]; the implementation specifics are part of our future work. In the SBP execution phase, the SBP engine publishes events which are received and evaluated by the SBP monitor to calculate the KPI values and display them in dashboards (Figure 6). Instance metrics are evaluated based on incoming events, which the SBP Monitor subscribes to. An event contains the identifier of the corresponding process instance.

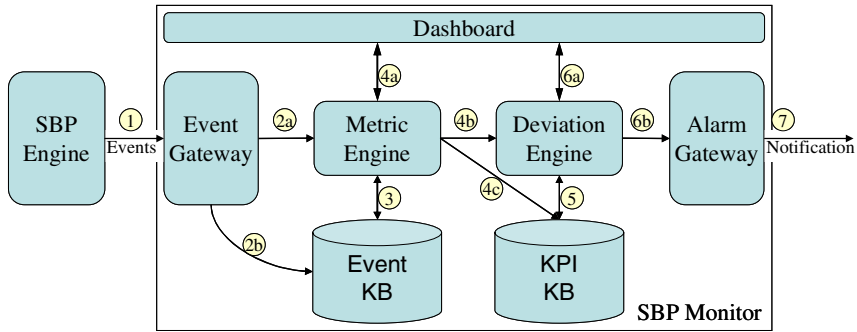


Fig. 6. SBP Monitor Architecture

When the process monitor receives an event, the event and corresponding data (business objects) that the event contains, are inserted into the knowledge base (KB) of the corresponding process instance. The metric engine determines in which instance metrics the event is used, as more than one instance metric could use this event. To evaluate a state metric, the condition of the state metric which is formulated as a logical expression acts as a query to the reasoner. In our example, the reasoner calculates whether the order was delivered on time, checks whether “OrderDelivered-Successfully(?order)” holds, and infers whether all items of the order are electronic products. For the evaluation of a duration metric, it has to be waited until the second event with the end date is received, until the computation can be performed. After the evaluation of the instance metrics all other dependent metrics are evaluated, i.e. instance metrics are evaluated first; followed by composite instance metrics; aggregate metrics are updated every time a new aggregated instance metric has been calculated; finally, composite aggregate metrics are computed. When the corresponding aggregate metric has been computed, the value of the KPI is updated in the KPI KB. After every update of the KPI’s actual value, the deviation rules are checked. In case of a deviation from the allowed value ranges, the deviation engine sends a notification to a business user (e.g. an email), shows a message in the dashboard, or invokes a web service.

5 Related Work

There are several approaches to performance management based on KPIs and BAM described in the literature and already implemented in products. IBM’s approach integrates performance management tightly into the BP lifecycle in a similar manner as we do, and supports it through its WebSphere family of products [WAM+07] in a model-driven way [CJ06]. However, the computation formula has to be specified by IT developers in the workflow layer. In our approach the KPI is completely defined by the business analyst based on the semantic annotation of the business process

model. In [CCD+04], the HP Business Process Cockpit product is described which supports KPI definition by business analysts based on so called behavior templates (similar to the KPI templates shown in Figure 2). However, behavior templates have to be predefined by IT engineers using SQL queries over the audit trail. None of these approaches uses ontologies for the description of business objects, and thus does not support inferring of implicit knowledge during metric evaluation.

In [APA+07] the vision of semantic process mining and monitoring is outlined. Semantic process mining deals with analysis of already executed process instances; process monitoring analyzes executing process instances. For SBP Monitoring a 5-phases “Observe-Evaluate-Detect-Diagnose-Resolve” approach is suggested. In this paper we deal with the first three phases: observation of events, evaluation of KPIs, and detection of deviations and resulting alerts.

The BAM approach described in this paper should be differentiated from other forms of business process analysis. BAM denotes real-time event-based monitoring, usage of dashboards to display business-level metrics to business decision makers, and automatic notification in case of deviations. This kind of monitoring is also called “active monitoring” as opposed to “passive monitoring” [MR00]. In passive monitoring the events and metrics are not actively evaluated by the system, but are explicitly issued by the user. The user can do this e.g. by ad-hoc querying the audit trail or asking the process engine about information on the running process instances. Completely other analysis techniques are process mining and data mining [CCD+04] based approaches. These techniques go beyond just reporting what is happening in the processes, but try to find out the causes and predict future behavior.

6 Conclusions and Outlook

In this paper we have presented an approach to semantic business process performance management based on key performance indicators.⁵ Compared to existing work, the benefit of our approach is (i) that KPIs can be fully specified by business analysts in the process modeling phase, because the needed information, namely business objects and their state changes, is available as semantic annotations of activities; (ii) as KPIs are specified based on business objects which are defined in ontologies, machine reasoning can exploit implicit knowledge during their evaluation. While the former aspect could also be implemented in non-semantic approaches, because it only requires that business objects (not necessarily defined in ontologies) are part of annotations of process models, the latter aspect is constrained to the use of ontologies.

In our future work we will implement the described approach by extending an existing SBP modeling tool and building an SBP monitor. In addition we will extend the KPI ontology by adding the possibility to specify process cost based KPIs and KPIs which are based on more than one business process. Apart from real-time active monitoring, we will investigate how semantics can be used for ad-hoc queries over audit trails (passive monitoring).

⁵ This work has in part been funded through the European Union's 6th Framework Programme, within Information Society Technologies (IST) priority under the SUPER project (FP6-026850, <http://www.ip-super.org>).

References

- [APA+07] Alves de Medeiros, A.K., Pedrinaci, C., Aalst, W.M.P., Domingue, J., Song, M., Rozinat, A., Norton, B., Cabral, L.: An Outlook on Semantic Business Process Mining and Monitoring. In: Proceedings of 3rd International IFIP Workshop On Semantic Web & Web Semantics (SWWS 2007) at On The Move Federated Conferences and Workshops (2007)
- [BDW07] Born, M., Doerr, F., Weber, I.: User-friendly Semantic Annotation in Business Process Modeling. In: Benatallah, B., Casati, F., Georgakopoulos, D., Bartolini, C., Sadiq, W., Godart, C. (eds.) WISE 2007. LNCS, vol. 4831, Springer, Heidelberg (2007)
- [BLK+05] de Bruijn, J., Lausen, H., Krummenacher, R., Polleres, A., Predoiu, L., Kifer, M., Fensel, D.: The Web Service Modeling Language WSML (October 5, 2005), <http://www.wsmo.org/TR/d16/d16.1/v0.21/>
- [BPMN06] Business Process Modeling Notation Specification. OMG Final Adopted Specification, February 6 (2006)
- [CCD+04] Castellanos, M., Casati, F., Dayal, U., Shan, M.-C.: A Comprehensive and Automated Approach to Intelligent Business Processes Execution Analysis. Distributed and Parallel Databases Journal 16(3) (2004)
- [CJ06] Chowdhary, P., Jeng, J.J., et al.: Model Driven Development for Business Performance Management. IBM System Journal 45(3) (2006)
- [Fa07] Fantini, P. (ed.): Semantic Business Process Lifecycle. SUPER Project Deliverable D2.2 (May 2007), <http://www.ip-super.org/res/Deliverables/M12/D2.2.pdf>
- [He07] Heymans, S. (ed.): Business Process Ontology Framework. SUPER Project Deliverable D1.1 (May 2007), <http://www.ip-super.org/res/Deliverables/M12/D1.1.pdf>
- [HLD+05] Hepp, M., Leymann, F., Domingue, J., Wahler, A., Fensel, D.: Semantic Business Process Management: A Vision Towards Using Semantic Web Services for Business Process Management. In: Proceedings of the IEEE ICEBE 2005, Beijing, China, October 18-20, 2005, pp. 535–540 (2005)
- [Lu02] Luckham, D.C.: The Power of Events: An Introduction to Complex Event Processing in Distributed Enterprise Systems. Addison-Wesley Longman Publishing Co., Inc., Boston, MA, USA (2002)
- [MR00] zur Muehlen, M., Rosemann, M.: Workflow-based Process Monitoring and Controlling - Technical and Organizational Issues. In: HICSS 2000 (2000)
- [NLK+07] Nitzsche, J., van Lessen, T., Karastoyanova, D., Leymann, F.: BPEL for Semantic Web Services. In: Proceedings of the 3rd International Workshop on Agents and Web Services in Distributed Environments (AWeSome 2007) (2007)
- [PD07] Pedrinaci, C., Domingue, J.: Towards an Ontology for Process Monitoring and Mining. In: Semantic Business Process and Product Lifecycle Management (SBPM 2007), Innsbruck, Austria (June 2007)
- [SCOR07] Supply Chain Operations Reference Model, SCOR Version 8.0 (2007), <http://www.supply-chain.org>
- [SF03] Smith, H., Fingar, P.: Business Process Management. The Third Wave. Meghan-Kiffer, US (2003)
- [SS06] Schmelzer, H.J., Sesselmann, W.: Geschäftsprozessmanagement in der Praxis. Hanser Verlag (2006)
- [WAM+07] Wahli, U., Avula, V., Macleod, H., Saeed, M., Vinther, A.: Business Process Management: Modeling through Monitoring Using WebSphere V6.0.2 Products. IBM Redbook SG24714801 (2007)